

Research Notes on Linear discriminant analysis

Linear discriminant analysis

>> Section 1

Without any further assumptions, the resulting classifier is referred to as quadratic discriminant analysis (QDA). LDA instead makes the additional simplifying homoscedasticity assumption (i.e. that the class covariances are identical, so $\Sigma_0 = \Sigma_1 = \Sigma$ $\{\displaystyle \Sigma_0 = \Sigma_1 = \Sigma\}$) and that the covariances have full rank. In this case, several terms cancel:

>> Section 2

Comparison to logistic regression Discriminant function analysis is very similar to logistic regression, and both can be used to answer the same research questions. Logistic regression does not have as many assumptions and restrictions as discriminant analysis. However, when discriminant analysis' assumptions are met, it is more powerful than logistic regression. Unlike logistic regression, discriminant analysis can be used with small sample sizes. It has been shown that when sample sizes are equal, and homogeneity of variance/covariance holds, discriminant analysis is more accurate. Despite all these advantages, logistic regression has none-the-less become the common choice, since the assumptions of discriminant analysis are rarely met.

>> Section 3

This means that the criterion of an input $x \rightarrow \{\displaystyle \vec{x}\}$ being in a class y $\{\displaystyle y\}$ is purely a function of this linear combination of the known observations. It is often useful to see this conclusion in geometrical terms: the criterion of an input $x \rightarrow \{\displaystyle \vec{x}\}$ being in a class y $\{\displaystyle y\}$ is purely a function of projection of multidimensional-space point $x \rightarrow \{\displaystyle \vec{x}\}$ onto vector $w \rightarrow \{\displaystyle \vec{w}\}$ (thus, we only consider its direction). In other words, the observation belongs to y $\{\displaystyle y\}$ if corresponding $x \rightarrow \{\displaystyle \vec{x}\}$ is located on a certain side of a hyperplane perpendicular to $w \rightarrow \{\displaystyle \vec{w}\}$. The location of the plane is defined by the threshold c $\{\displaystyle c\}$.

>> Section 4

Effect size Some suggest the use of eigenvalues as effect size measures, however, this is generally not supported. Instead, the canonical correlation is the preferred measure of effect size. It is similar to the eigenvalue, but is the square root of the ratio of SS_{between} and SS_{total} . It is the correlation between groups and the function. Another popular measure of effect size is the percent of variance for each function. This is calculated by: $(\lambda \times \sum_i \lambda_i) \times 100$ $\{\displaystyle \left(\frac{\lambda_x}{\sum_i \lambda_i}\right) \times 100\}$ where λ_x $\{\displaystyle \lambda_x\}$ is the eigenvalue for the function and $\sum_i \lambda_i$ $\{\displaystyle \sum_i \lambda_i\}$ is the sum of all eigenvalues. This tells us how strong the prediction is for that particular function compared to the others. Percent correctly classified can also be analyzed as an effect size. The kappa value can describe this while correcting for chance agreement. Kappa normalizes across all categorizes rather than biased by a significantly good or poorly performing classes.

>> Section 5

Assumptions The assumptions of discriminant analysis are the same as those for MANOVA. The analysis is quite sensitive to outliers and the size of the smallest group must be larger than the number of predictor variables.

>> Section 6

Linear discriminant analysis (LDA), normal discriminant analysis (NDA), canonical variates analysis (CVA), or discriminant function analysis is a generalization of Fisher's linear discriminant, a method used in statistics and other fields, to find a linear combination of features that characterizes or separates two or more classes of objects or events. The resulting combination may be used as a linear classifier, or, more commonly, for dimensionality reduction before later classification. LDA is closely related to analysis of variance (ANOVA) and regression analysis, which also attempt to express one dependent variable as a linear combination of other features or measurements. However, ANOVA uses categorical independent variables and a continuous dependent variable, whereas discriminant analysis has continuous independent variables and a categorical dependent variable (i.e. the class label). Logistic regression and probit regression are more similar to LDA than ANOVA is, as they also explain a categorical variable by the values of continuous independent variables. These other methods are preferable in applications where it is not reasonable to assume that the independent variables have a normal distribution, which is a fundamental assumption of the LDA method. LDA is also closely related to principal component analysis (PCA) and factor analysis in that they both look for linear combinations of variables which best explain the data. LDA explicitly attempts to model the difference between the classes of data. PCA, in contrast, does not take into account any difference in class, and factor analysis builds the feature combinations based on similarities rather than differences. Discriminant analysis is also different from factor analysis in that it is not an interdependence technique: a distinction between independent variables and dependent variables (also called criterion variables) must be made. LDA works when

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the measurements made on independent variables for each observation are continuous quantities. When dealing with categorical independent variables, the equivalent technique is discriminant correspondence analysis. Discriminant analysis is used when groups are known a priori (unlike in cluster analysis). Each case must have a score on one or more quantitative predictor measures, and a score on a group measure. In simple terms, discriminant function analysis is classification - the act of distributing things into groups, classes or categories of the same type.

>> Section 7

Discriminant Correlation Analysis (DCA) of the Haghighat article (see above) ALGLIB contains open-source LDA implementation in C# / C++ / Pascal / VBA. LDA in Python- LDA implementation in Python LDA tutorial using MS Excel Biomedical statistics. Discriminant analysis StatQuest: Linear Discriminant Analysis (LDA) clearly explained on YouTube Course notes, Discriminant function analysis by G. David Garson, NC State University Discriminant analysis tutorial in Microsoft Excel by Kardi Teknomo Course notes, Discriminant function analysis by David W. Stockburger, Missouri State University Archived 2016-03-03 at the Wayback Machine Discriminant function analysis (DA) by John Poulsen and Aaron French, San Francisco State University Archived 2011-12-15 at the Wayback Machine

>> Section 8

Multivariate normality: Independent variables are normal for each level of the grouping variable. Homogeneity of variance/covariance (homoscedasticity): Variances among group variables are the same across levels of predictors. Can be tested with Box's M statistic. It has been suggested, however, that linear discriminant analysis be used when covariances are equal, and that quadratic discriminant analysis may be used when covariances are not equal. Independence: Participants are assumed to be randomly sampled, and a participant's score on one variable is assumed to be independent of scores on that variable for all other participants. It has been suggested that discriminant analysis is relatively robust to slight violations of these assumptions, and it has also been shown that discriminant analysis may still be reliable when using dichotomous variables (where multivariate normality is often violated).